CONGRUENCE OF ANGLES AND TRIANGLES

6–1 INTRODUCTION

In the last two chapters we have been concerned with congruence of segments only. But we also have a feeling for size of an angle—and, in fact, in Chapter 1 we used a protractor to measure angles. If our geometry is to be like the world of our experience, we must include in it postulates that will enable us to compare the sizes of angles. These postulates form Postulate Group V. There are four postulates in the group, and we take them up one at a time. As in Chapter 4, we shall use the word "congruent" to give the idea of *the same size*.

6–2 POSTULATE V–1, THE EXISTENCE POSTULATE FOR ANGLES

This first postulate of Group V tells of the existence of angles congruent to a given angle. Remember to think of the word "congruent" as meaning *the same size*.

V–1 If $\angle A$ is not a straight angle, and if r' is a ray from A' on a line l', then on a given side of l', there is exactly one ray s' from A' such that the angle A', with sides r' and s', is congruent to angle A (Fig. 6–1). We write this as $\angle A' \cong \angle A$. If $\angle A$ is a straight angle, then the straight angle at A', with one side r', is the only angle with one side r' congruent to angle A.

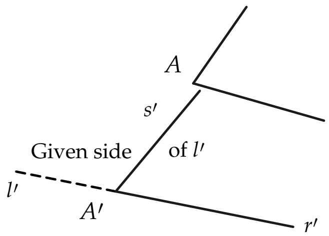


FIGURE 6–1

Exercise Group 6–1

1. If $\angle A$ is not a straight angle, how many angles with side r' are congruent to $\angle A$? How many if $\angle A$ is a straight angle?
2. In a figure like 6–2 draw the possible positions of the side s' such that $\angle A' \cong \angle A$.

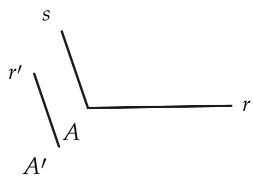


FIGURE 6–2

3. In a figure like 6–3, draw all possible positions of s' such that $\angle A' \cong \angle A$.

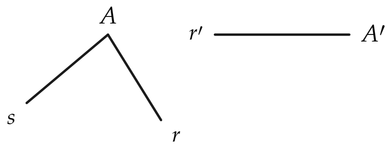


FIGURE 6–3

4. If in Fig. 6–4, A is a straight angle and r' is a ray from A', draw an angle, with one side r', that is congruent to A. Is it the only such angle?

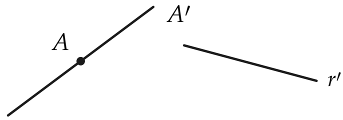


FIGURE 6–4

5. If $\angle O$ is a straight angle and $\angle O' \cong \angle O$, is $\angle O'$ a straight angle? Why?

6–3 POSTULATE V–2, THE THREE-PART ANGLE POSTULATE

The next postulate is for angles what Postulate III–2 is for segments. It may seem odd to have to state such seemingly obvious things, but remember we are dealing with those imaginary things called points and lines, and so we must state *all* the properties we need.

V–2 (a) $\angle A \cong \angle A$. (Congruence of angles is reflexive.)
(b) If $\angle A \cong \angle B$, then $\angle B \cong \angle A$. (Congruence of angles is symmetric.)
(c) If $\angle A \cong \angle B$, and $\angle B \cong \angle C$, then $\angle A \cong \angle C$. (Congruence of angles is transitive.)

Exercise Group 6–2

1. State each part of Postulate V–2 in words.
2. If $\angle E \cong \angle F$, and $\angle F \cong \angle G$, and $\angle G \cong \angle H$, show, by using part (c) of Postulate V–2, that $\angle E \cong \angle H$.
3. If $\angle R \cong \angle S$ and $\angle T \cong \angle S$, prove that $\angle R \cong \angle T$.

*4. Use Postulate V–1 and (b) and (c) of Postulate V–2 to prove that for every $\angle A$, $\angle A \cong \angle A$.

6–4 POSTULATE V–3, THE ADDITION AND SUBTRACTION OF ANGLES POSTULATE

The next postulate does for angles what Postulate III–3 does for segments. It enables us to "add" and "subtract" angles. To avoid a complicated statement in words, we explain the postulate by using Fig. 6–5, where the side common to $\angle B$ and $\angle C$, except the vertex, is in the interior of $\angle A$, and the same for $\angle A'$, $\angle B'$, and $\angle C'$.

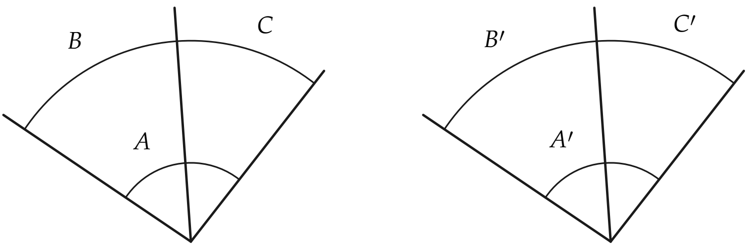


FIGURE 6–5

V–3 If the angles in any two of the pairs $\angle A$ and $\angle A'$, $\angle B$ and $\angle B'$, $\angle C$ and $\angle C'$ are congruent to each other, then the angles of the third pair are also congruent to each other. In other words:

(a) If $\angle A \cong \angle A'$ and $\angle B \cong \angle B'$, then $\angle C \cong \angle C'$.

(b) If $\angle B \cong \angle B'$ and $\angle C \cong \angle C'$, then $\angle A \cong \angle A'$.

(c) If $\angle C \cong \angle C'$ and $\angle A \cong \angle A'$, then $\angle B \cong \angle B'$.

The angle A may be a straight angle.

Exercise Group 6–3

1. We have not yet in our geometry studied the measurement of angles in degrees. However, in this exercise, use the fact (which will be proved later) that angles congruent to each other have the same measure in degrees, and add or subtract degree measures as you are accustomed to do. Refer to Fig. 6–6.
   (a) If $\angle A$ is 130° and $\angle B$ is 45°, how many degrees in $\angle C$?
   (b) If $\angle B'$ is 41° and $\angle C'$ is 75°, how many degrees in $\angle A'$?
   (c) If $\angle A \cong \angle A'$ and $\angle A$ is 150°, how many degrees in $\angle A'$?
   (d) If $\angle A \cong \angle A'$ and $\angle C \cong \angle C'$, are $\angle B$ and $\angle B'$ congruent to each other?
   (e) If $\angle A$ is 87° and congruent to $\angle A'$, and if $\angle C \cong \angle C'$ and $\angle C'$ is 33°, how many degrees in $\angle B$?

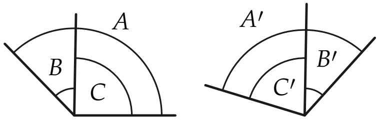


FIGURE 6–6

2. In Fig. 6–7, if $\angle POR \cong \angle SVU$, and $\angle QOR \cong \angle TVU$, what can you say about $\angle POQ$?

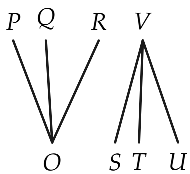


FIGURE 6–7

3. In Fig. 6–8, if $\angle BAC \cong \angle DAE$, what can you say about $\angle BAD$?

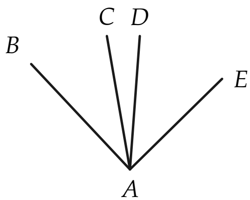


FIGURE 6–8

4. In Fig. 6–9, if $\angle SPU \cong$

$\angle RPT$, what can you say about $\angle RPS$?

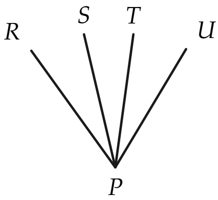


FIGURE 6–9

5. Is Exercise 2 an example of "adding" or "subtracting" angles? What about Exercises 3 and 4?

Supplementary and adjacent angles were defined in Definition 3–9. The statement that $\angle A$ is *supplementary* to $\angle B$ shall mean that $\angle A$ is congruent to an angle that is *supplementary and adjacent* to $\angle B$.

Theorem 6–1. *If two angles are congruent to each other, then their supplements are also congruent to each other.*

Proof. Let the given angles be A and A', and let their supplements be B and B', respectively, as in Fig. 6–10. We are given that $\angle A \cong \angle A'$. Since any two straight angles are congruent to each other, it follows at once from Postulate V–3 that $\angle B \cong \angle B'$. (Why?)

FIGURE 6–10

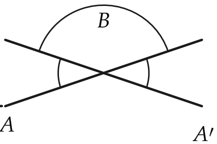


FIGURE 6–11

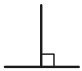


FIGURE 6–12

Theorem 6–2. *Angles that are vertical to each other are also congruent to each other.*

Proof. Let the angles A and A' be vertical to each other, as in Fig. 6–11, and consider $\angle B$, one of the other angles formed by the lines. Then $\angle B$ is the supplement of $\angle A$ and also the supplement of $\angle A'$. Since $\angle B \cong \angle B$, it follows from Theorem 6–1 that $\angle A \cong \angle A'$.

Definition 6–1. An angle (Fig. 6–12) that is congruent to its supplement is called a *right angle*.

Remark. Note that the words *congruent*, *vertical*, *equal*, and *supplementary* refer to relations between two objects, not to their properties. That is, it makes no sense to say that $\angle A$ is a vertical angle, or a congruent angle, or a supplementary angle. At times, we refer to two angles A and B as equal angles, or vertical angles, or congruent angles, etc. What we really mean is that $\angle A$ is equal to $\angle B$, $\angle A$ is vertical to $\angle B$, $\angle A$ is congruent to $\angle B$, etc. It is important to distinguish between properties and relations.

Exercise Group 6–4

1. If an angle is a right angle, what is its supplement?
2. If $\angle 1 \cong \angle 2$ in Fig. 6–13, prove that $\angle EAC \cong \angle DAB$.

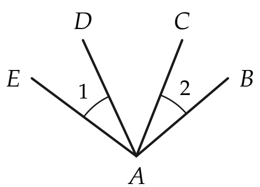


FIGURE 6–13

3. In Fig. 6–14 it is given that $\angle B$ is congruent to the supplement of $\angle C$ and $\angle C \cong \angle D$. Prove that $\angle A \cong \angle E$.

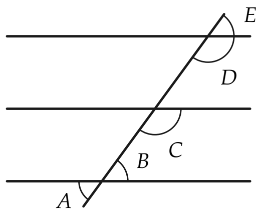


FIGURE 6–14

*4. An angle is congruent to a

right angle. Show that it, too, is a right angle.

5. In Fig. 6–15, the line CD bisects $\angle AOB$. Show that it also bisects $\angle EOF$. (By "bisect" we mean that $\angle AOC \cong \angle COB$.)

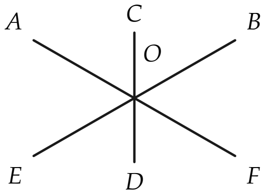


FIGURE 6–15

6. Decide whether the following words refer to properties or relations: perpendicular, parallel, vertex, altitude, greater than, angle, segment, median, right (angle), straight (angle), bisector.

6–5 POSTULATE V–4, THE SIDE-ANGLE-SIDE POSTULATE

This postulate (abbreviated S.A.S.) is one that combines congruence of segments with congruence of angles. Up to this point, we have the two kinds of congruence but no relationship between them. This is the connecting link that will enable us to prove many theorems.

V–4 If two triangles, $\triangle ABC$ and $\triangle A'B'C'$, are such that $AB \cong A'B'$, $AC \cong A'C'$, and $\angle A \cong \angle A'$, then $\angle B \cong \angle B'$ and $\angle C \cong \angle C'$ (Fig. 6–16).

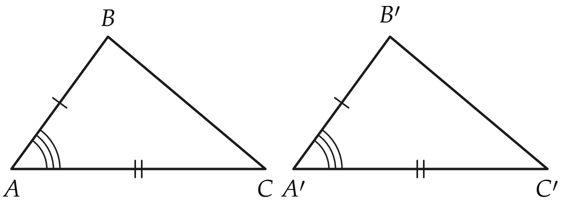


FIGURE 6–16

This postulate tells us that if two triangles have two sides of one congruent to two sides of the other, and if the angles formed by these two sides (called the *included* angles) are congruent to each other, then the other pairs of corresponding angles are congruent.

Exercise Group 6–5

1. In Fig. 6–17, if $AB \cong DF$, $BC \cong DE$, and $\angle B \cong \angle D$, what angle is congruent to $\angle A$? to $\angle C$?

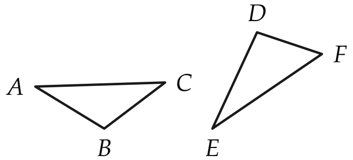


FIGURE 6–17

2. In Fig. 6–18, if $PQ = US$, $QR = UT$, and $\angle Q \cong \angle U$, what congruences hold?

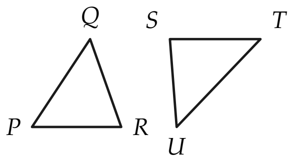


FIGURE 6–18

3. Draw figures to show that if $AB \cong A'B'$, and $\angle A \cong \angle A'$, but $AC \ncong A'C'$ (the symbol $\ncong$ is read "is not congruent to"), then it need not be true that $\angle C \cong \angle C'$.
4. Draw figures to show that if $AB \cong A'B'$, and $AC \cong A'C'$, but $\angle A \ncong \angle A'$, then it need not be true that $\angle B \cong \angle B'$.
5. If, in the figure $AC \cong AD$, and $\angle CAB \cong \angle BAD$, prove that $\angle C \cong \angle D$. (Fig. 6–19)

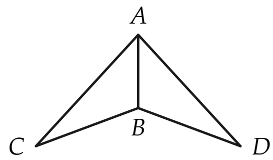


FIGURE 6–19

6. If, in the figure, $DO \cong OB$, and $AO \cong OC$, prove that $\angle A \cong \angle C$. (Fig. 6–20)

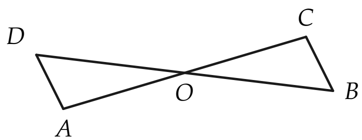


FIGURE 6–20

7. If, in the figure, $AB \cong CD$, and $\angle CBA \cong \angle BCD$, prove that $\angle CAB \cong \angle BDC$. (Fig. 6–21)

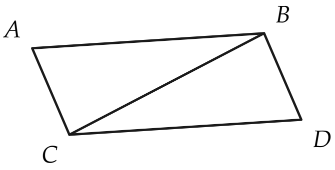


FIGURE 6–21

8. If, in the figure, $AD \cong BD$, and $\angle ADC$ is a right angle, prove that $\angle A \cong \angle B$. (Fig. 6–22)

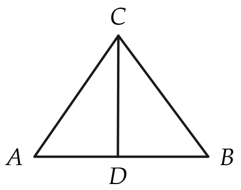


FIGURE 6–22

9. Given: $\angle EBC \cong \angle ABD$,
   $AB \cong BC$,
   $BE \cong BD$.
   Prove: $\angle A \cong \angle C$.
   (Fig. 6–23)

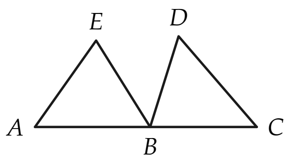


FIGURE 6–23

10. Given: $AD \cong DC$,
    $\angle 1 \cong \angle C$,
    $\angle 2 \cong \angle C$.
    Prove: $\angle 3 \cong \angle 4$.
    (Fig. 6–24)

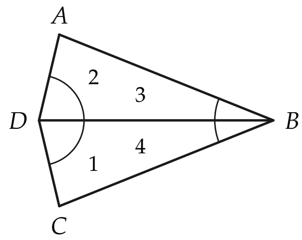


FIGURE 6–24

11. Given: $\angle 1 \cong \angle 2$,
    $\angle BCF \cong \angle ADF$,
    $CG \cong HD$,
    $CF \cong DF$.
    Prove: $\angle 3 \cong \angle 4$.
    (Fig. 6–25)

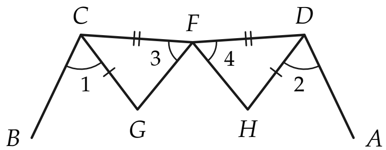


FIGURE 6–25

12. Given: $\angle 1 \cong \angle 2$,
    $AE \cong CD$,
    $AB \cong BC$.
    Prove: $\angle E \cong \angle D$.
    (Fig. 6–26)

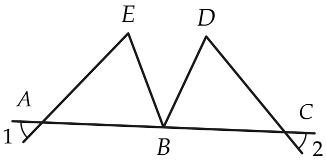


FIGURE 6–26

13. Given: $\angle 1 \cong \angle 2$,
$AC \cong BC$,
$AM \cong BM$.
Prove: $\angle 3 \cong \angle 4$.
(Fig. 6–27)

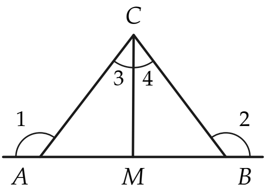


FIGURE 6–27

14. Given: $\angle ADC \cong \angle ABC$,
$\angle 1 \cong \angle 2$,
$AD \cong BC$.
Prove: $\angle A \cong \angle C$.
(Fig. 6–28)

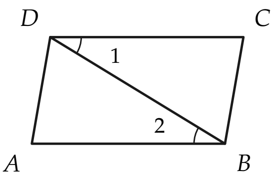


FIGURE 6–28

15. Given: $AD \cong DC$,
$\angle 1 \cong \angle 2$.
Prove: $\angle A \cong \angle C$.
(Fig. 6–29)

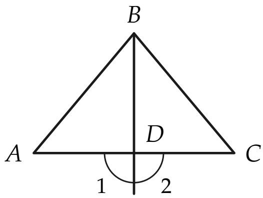


FIGURE 6–29

*16. Given: $BF \cong ED$,
$AE \cong CF$,
$\angle 1 \cong \angle 2$.
Prove: $\angle 3 \cong \angle 4$,
$\angle 5 \cong \angle 6$.
(Fig. 6–30)

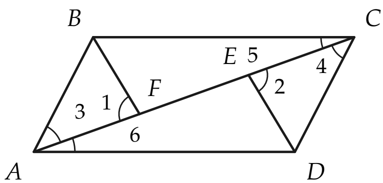


FIGURE 6–30

*17. Given: $AB \cong AD$,
$BO \cong OD$,
$\angle OAB \cong \angle OAD$.
Prove: $\angle 1 \cong \angle 2$.
(Fig. 6–31)

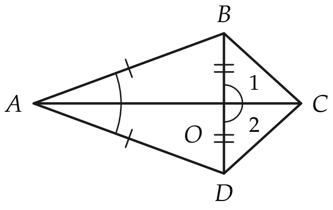


FIGURE 6–31

6–6 ISOSCELES TRIANGLES

We can now prove a theorem about isosceles triangles, for which we need a definition. The definition of isosceles triangle is the same as that given in Chapter 1, Exercise Group 1–4:

Definition 6–2. A triangle is called *isosceles* if two of its sides are congruent (of equal length). The *base angles* of the triangle are those opposite the congruent sides.

Theorem 6–3. *In every isosceles triangle, the base angles are congruent to each other.*

Proof. Consider any isosceles triangle, and label the vertices in two different ways, as shown in Fig. 6–32. Let $AB \cong AC$. Let us look at the triangle both as $\triangle ABC$ and as $\triangle A'B'C'$. We have $AB \cong AC \cong A'B'$, and $AC \cong AB = A'C'$, and $\angle A \cong \angle A'$. Therefore, by the S.A.S. postulate, $\angle B \cong \angle B'$, that is, $\angle B \cong \angle C$.

This proof is quite interesting because it illustrates very well how good notation can help. In the proof we look at the same triangle in two different ways, and once we have done this, the proof is very short.

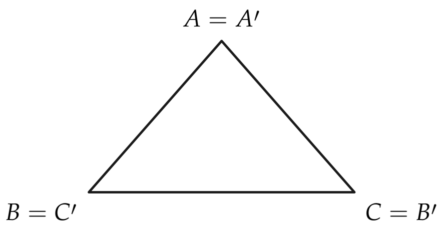


FIGURE 6–32

Definition 6–3. A *midpoint* of a segment AB is a point M of the segment, such that $AM \cong MB$.

Exercise Group 6–6

1. Using Theorem 6–3, prove that an equilateral triangle is equiangular.
2. If $PQ = PR$, and $\angle Q$ is 44°, how many degrees in $\angle R$? (We have not yet studied degrees in our geometry, but for this problem you may use what you already know from Chapter 1.) (Fig. 6–33)

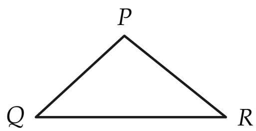


FIGURE 6–33

3. If, in the figure, $AB \cong AC$ and $BD \cong DC$, prove that

$$\angle ABC \cong \angle ACB,$$
$$\angle CBD \cong \angle BCD,$$
$$\angle ABD \cong \angle ACD.$$

(Fig. 6–34)

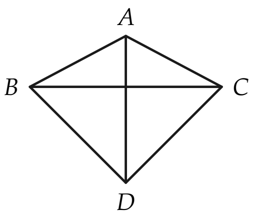


FIGURE 6–34

4. Given: $AC \cong BC$,
   $AD \cong DB$.
   Prove: $\angle 1 \cong \angle 2$.
   (Fig. 6–35)

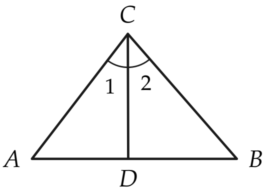


FIGURE 6–35

5. Given: $\angle ABC \cong \angle ABE$,
   $CB \cong BE$.
   Prove: $\angle C \cong \angle E$.
   (Fig. 6–36)

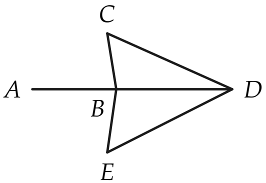


FIGURE 6–36

6. Given: $AC \cong DF$,
   $BD \cong CE$,
   $\angle 1 \cong \angle 2$.
   Prove: $\angle A \cong \angle F$.
   (Fig. 6–37)

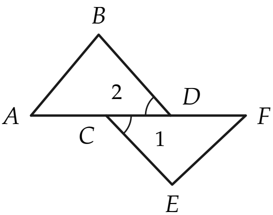


FIGURE 6–37

7. In the figure $AD \cong EC$, and $BE \cong BD$. Show that $\angle A \cong \angle C$. (Fig. 6–38)

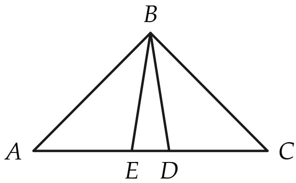


FIGURE 6–38

8. Given: $RS \cong RT$,
   $US \cong UT$.
   Prove: $\angle RSU \cong \angle RTU$.
   (Fig. 6–39)

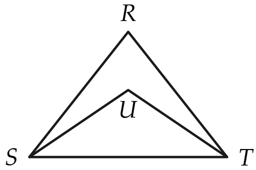


FIGURE 6–39

9. Given: $AC \cong BC$,
   $CD \cong CE$.
   Prove: $\angle 1 \cong \angle 2$.
   (Fig. 6–40)

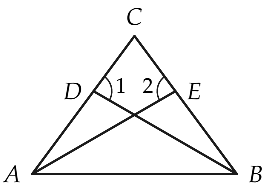


FIGURE 6–40

10. Given: $AO \cong OD$,
$AF \cong FD$,
$BO \cong OC$,
$BF \cong FC$.
Prove: $\angle EAB \cong \angle EDC$. (Fig. 6–41)

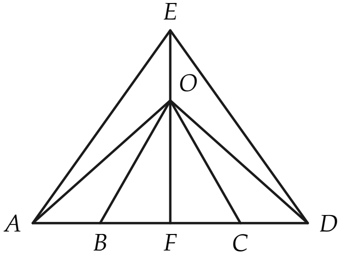


FIGURE 6–41

11. In the figure, $AC \cong BC$, $CG \cong CF$, and $\angle 1 \cong \angle 2$. Prove that $\angle 3 \cong \angle 4$. (Fig. 6–42)

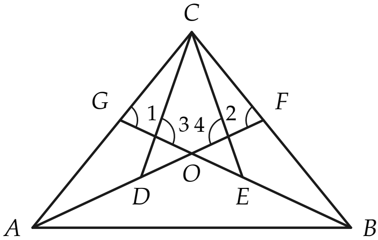


FIGURE 6–42

6–7 CONGRUENCE OF TRIANGLES

We now define congruence for triangles and prove some theorems.

Definition 6–4. Two triangles are said to be *congruent* to each other if they can be labeled $\triangle ABC$ and $\triangle A'B'C'$, such that $AB \cong A'B'$, $AC \cong A'C'$, $BC \cong B'C'$, $\angle A \cong \angle A'$, $\angle B \cong \angle B'$, and $\angle C \cong \angle C'$. We then write $\triangle ABC \cong \triangle A'B'C'$ and read this as "triangle ABC is congruent to triangle $A'B'C'$." $\angle A$ corresponds to $\angle A'$, $\angle B$ to $\angle B'$, side AC to side $A'C'$, etc.

Exercise 6–7

Prove that congruence of triangles is transitive.

Theorem 6–4. *If the vertices of two triangles may be labeled A, B, C and A', B', C', such that $AB \cong A'B'$, $AC \cong A'C'$, $\angle A \cong \angle A'$, then $\triangle ABC \cong \triangle A'B'C'$.*

Remark. This is the first of the three congruence theorems for triangles. It may also be stated as follows: *if two triangles are such that two sides and the included angle of one are congruent, respectively, to two sides and the included angle of the other, then the triangles are congruent.* This theorem is often called the "side-angle-side theorem" and is abbreviated S.A.S.

Proof. From the S.A.S. postulate, we know that $\angle B \cong \angle B'$ and $\angle C \cong \angle C'$. It only remains to be shown that the third sides are congruent, that is, that $BC \cong B'C'$. We prove this by considering the point D' on ray $C'B'$, such that $CB \cong C'D'$, as in Fig. 6–43, and then showing that $D' = B'$.

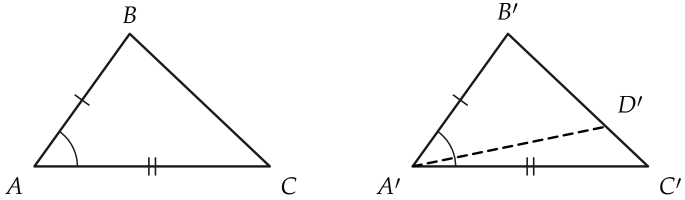


FIGURE 6–43

STATEMENTS	REASONS
1. There is a point D' on the ray $C'B'$ such that $CB \cong C'D'$.	Existence postulate for segments.
2. $AC \cong A'C'$, and $AB \cong A'B'$.	Given.
3. $\angle B'A'C' \cong \angle BAC$.	Given.
4. $\angle C \cong \angle C'$, and $\angle B \cong \angle B'$.	S.A.S. postulate.
5. $\angle BAC \cong \angle D'A'C'$.	By S.A.S. postulate, considering $\triangle ACB$ and $\triangle A'C'D'$.
6. Ray $A'B'$ is the same as the ray $A'D'$.	Existence postulate for angles.
7. $B' = D'$.	The lines $A'B'$ and $B'C'$ intersect in exactly one point by Theorem 3–1. This point is B' and is also D'.
8. $BC \cong B'C'$.	Statements 1 and 7.
9. $\triangle ABC \cong \triangle A'B'C'$.	Definition 6–4.

In the exercises that follow we will make use of the congruence theorems. In some exercises it will be convenient to know that *"halves" of congruent segments are congruent to each other*. That is, if $AB \cong CD$ and M and N are midpoints of AB and CD respectively, then $AM \cong CN$. To see this we note that $\overline{AB} = \overline{CD}$ and by the properties of numbers $\overline{AM} = \overline{CN}$ so that $AM \cong CN$.

Exercise Group 6–8

1. Given the figure, let $\angle 1 \cong \angle 2$ and $AB \cong BC$. Prove that $AD \cong DC$. (Fig. 6–44)

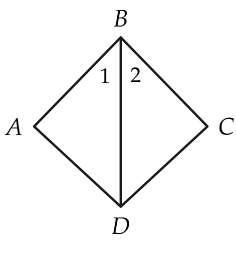


FIGURE 6–44

2. Given the figure, let $\angle 1 \cong \angle 2$ and $EF \cong GH$. Prove that $\triangle EGH = \triangle HFE$. (Fig. 6–45)

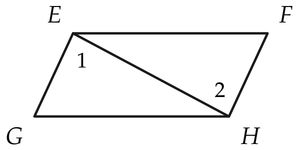


FIGURE 6–45

3. Given the figure, let $AC \cong AD$ and AB bisect $\angle A$. Prove that $\angle CBA \cong \angle DBA$. (Fig. 6–46)

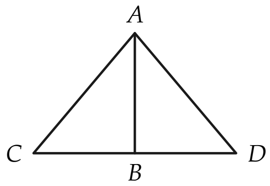


FIGURE 6–46

4. Given the figure, let $AB \cong AD$ and $BC \cong CD$. Prove that $\triangle ABC \cong \triangle ADC$. (Fig. 6–47)

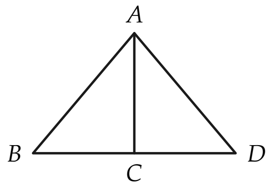


FIGURE 6–47

5. If, in the figure, $AB \cong AC$, D is the midpoint of AB, and E is the midpoint of AC, prove that $\triangle ABE \cong \triangle ACD$. (Fig. 6–48)

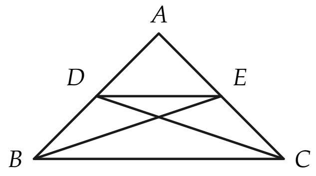


FIGURE 6–48

6. If, in the figure, $AB \cong AD$, and $BC \cong CD$, prove that $\angle ABC \cong \angle ADC$ and $\angle BAC \cong \angle DAC$. (Fig. 6–49)

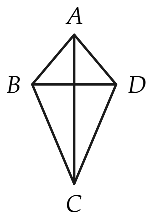


FIGURE 6–49

7. In the figure, $AB \cong BC$, and D, E are the midpoints of AB and BC respectively. Prove that $AE \cong CD$. (Fig. 6–50)

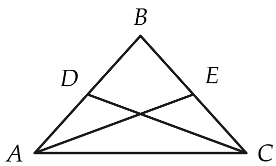


FIGURE 6–50

8. Given: $EB \cong BD$, B is the midpoint of AC, and $\angle EBC \cong \angle DBA$. Prove that $AE \cong CD$. (Fig. 6–51)

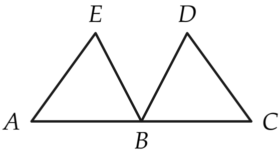


FIGURE 6–51

9. Prove that $BD \cong AC$ if $\angle DAB \cong \angle CBA$ and $AD \cong BC$. (Fig. 6–52)

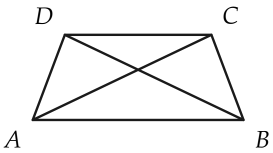


FIGURE 6–52

10. In the figure, $AD \cong EC$, and $BE \cong BD$. Prove that $\triangle ABC$ is isosceles. (Fig. 6–53)

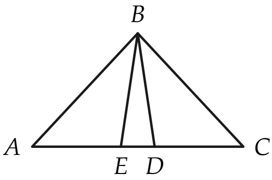


FIGURE 6–53

11. A farmer, wishing to find the distance between posts at A and B on opposite sides of a barn, does the following. He places a stake at C, and on the line through B and C places a stake at D, such that $\overline{BC} = \overline{DC}$. Then on the line through A and C, he places a stake at E, such that $\overline{AC} = \overline{CE}$. He measures DE and finds the length is 78 ft. Prove that $\overline{AB} = 78$ ft. (Fig. 6–54)

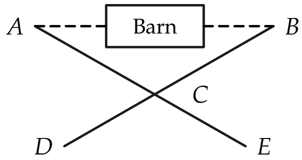


FIGURE 6–54

6–8 THE SECOND CONGRUENCE THEOREM

The first congruence theorem was the side-angle-side theorem (S.A.S.). The second one says that two triangles are congruent if two angles and the included side of one are, respectively, congruent to two angles and the included side of the other. This theorem is called the "angle-side-angle theorem" and is abbreviated A.S.A.

Theorem 6–5. *If two triangles $\triangle ABC$ and $\triangle A'B'C'$ are such that $\angle A \cong \angle A'$, $\angle C \cong \angle C'$, and $AC \cong A'C'$, then $\triangle ABC \cong \triangle A'B'C'$.*

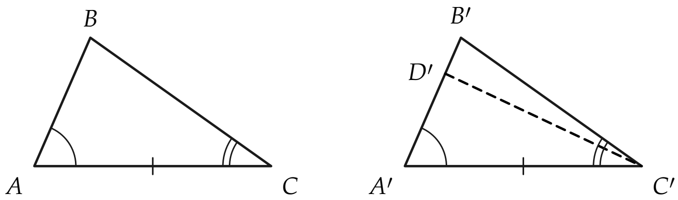


FIGURE 6–55

Proof. We consider the point D' on the ray $A'B'$ (Fig. 6–55), such that $AB \cong A'D'$. Then it is true that $\triangle ABC \cong \triangle A'D'C'$. (Why?) Then, if $D' \neq B'$, we have both $\angle D'C'A'$ and $\angle B'C'A'$ congruent to $\angle C$. This is impossible, so $B' = D'$.

STATEMENTS	REASONS
1. There is a point D' on the ray $A'B'$, such that $AB \cong A'D'$.	Existence postulate for segments.
2. $\angle A \cong \angle A'$, $AC \cong A'C'$.	Given.
3. $\angle C \cong \angle D'C'A'$.	S.A.S. postulate, applied to $\triangle ABC$ and $\triangle A'D'C'$.
4. $\angle C \cong \angle B'C'A'$.	Given.
5. Ray $C'D' =$ ray $C'B'$.	Existence postulate for angles.
6. $D' = B'$.	Two lines intersect in at most one point.
7. $\triangle ABC \cong \triangle A'B'C'$.	S.A.S. theorem.

Exercise Group 6–9

1. If, in the figure, $\angle 1 \cong \angle 2$, and $\angle 3 \cong \angle 4$, prove that $\triangle ABD \cong \triangle CBD$. (Fig. 6–56)

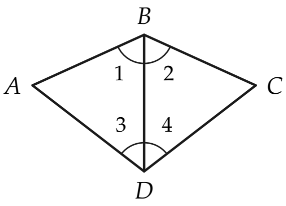


FIGURE 6–56

2. If, in the figure, $\angle 1 \cong \angle 2$, and $\angle 3 \cong \angle 4$, prove that $AB = CD$. (Fig. 6–57)

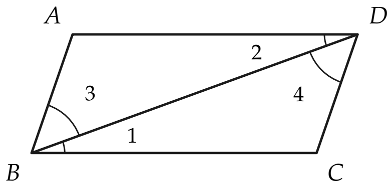


FIGURE 6–57

3. If, in the figure, $\angle 1 \cong \angle 2$, and $AB \cong BC$, prove that $AE \cong DC$. (Fig. 6–58)

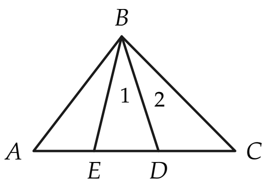


FIGURE 6–58

4. Draw a triangle. Let one angle be 31°, a second angle be 53°, and their included side be 7 inches.
5. If, in the figure, $\angle ABC \cong \angle ABE$, and $\angle CDB \cong \angle EDB$, prove that $CD \cong ED$. (Fig. 6–59)

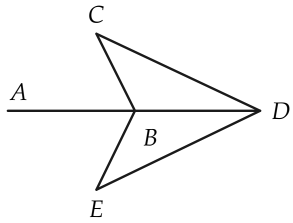


FIGURE 6–59

6. If, in the figure, $AB \cong BC$, and $\angle A \cong \angle C$, prove that $\triangle ABD \cong \triangle CBE$. (Fig. 6–60)

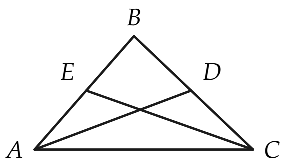


FIGURE 6–60

7. If, in the figure, $\angle A \cong \angle D$ and $AB \cong BD$, prove that $AE \cong CD$. (Fig. 6–61)

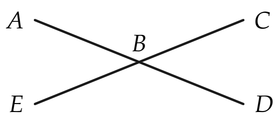


FIGURE 6–61

8. If, in the figure, $AF \cong CD$, $\angle A \cong \angle D$, and $\angle ACB \cong$

$\angle DFE$, prove that $\triangle ABC \cong \triangle DEF$. (Fig. 6–62)

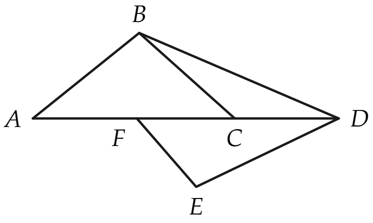


FIGURE 6–62

9. Draw triangles ABC whose parts are:
   (a) $\angle A = 36°$; $\angle B = 45°$; $AB = 3$ in.
   (b) $\angle A = 60°$; $\angle C = 30°$; $AC = 4$ in.
   (c) $\angle C = 33°$; $\angle B = 33°$; $BC = 5/2$ in.
10. Given: $\angle 3 \cong \angle 4$,
    $AB \cong AD$,
    AC bisects $\angle BAD$.
    Prove: $BC \cong CD$.
    (Fig. 6–63)

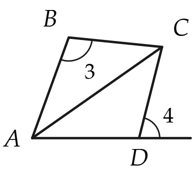


FIGURE 6–63

11. In the figure, $AF \cong FB$, and $AE \cong DB$. Prove that $\angle 3 \cong \angle 4$. (Fig. 6–64)

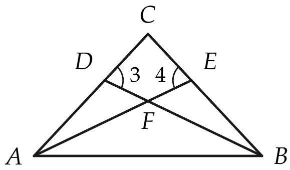


FIGURE 6–64

12. Prove that $\angle 1 \cong \angle 2$ if $AB \cong BC$ and $AD \cong DC$. (Fig. 6–65)

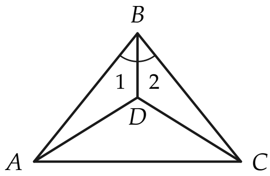


FIGURE 6–65

13. If $AH \cong HB$, and $AG \cong GB$, prove that $\angle 3 \cong \angle 4$. (Fig. 6–66)

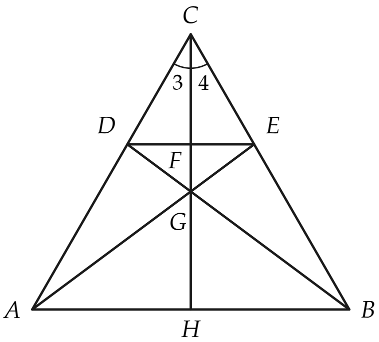


FIGURE 6–66

14. Given: $\angle 1 \cong \angle 2$,
$AC \cong BC$,
$DF \cong EG$.
Prove: $\triangle HIJ$ is isosceles.
(Fig. 6–67)

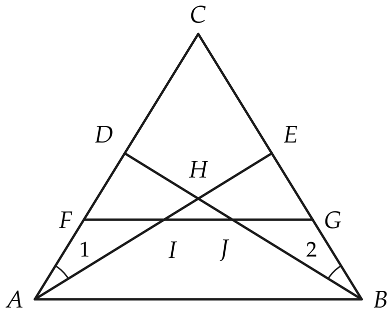


FIGURE 6–67

In Section 6–6 we proved that the base angles of an isosceles triangle are congruent. We can now easily prove the converse.

Theorem 6–6. *If a triangle has two angles congruent to each other, then the sides opposite these angles are congruent.*

Proof. Suppose that the triangle is $\triangle ABC$, and that $\angle A \cong \angle C$. Let the vertices be labeled A', B', C', as in Fig. 6–68. Then we have $\angle A \cong \angle C \cong \angle A'$, and $\angle C \cong \angle A \cong \angle C'$, and $AC \cong CA \cong A'C'$. Hence by the A.S.A. congruence theorem, $\triangle ABC \cong \triangle A'B'C'$. Therefore $AB \cong A'B' \cong CB$.

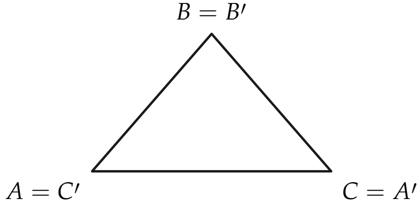


FIGURE 6–68

Exercise 6–10

Prove that an equiangular triangle is equilateral.

6–9 THE THIRD CONGRUENCE THEOREM

The two previous congruence theorems were the S.A.S. and A.S.A. theorems. The third of the three fundamental congruence theorems states that two triangles are congruent if the sides of one are congruent to the sides of the other. It is called the "side-side-side theorem" and is abbreviated S.S.S.

Theorem 6–7. *If the sides of one triangle are congruent, respectively, to the sides of another triangle, then the triangles are congruent.*

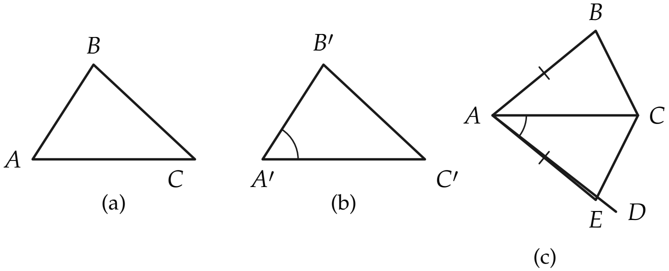


FIGURE 6–69

Proof. Suppose that the triangles are labeled $\triangle ABC$ and $\triangle A'B'C'$, as in Fig. 6–69(a) and 6–69(b), and that $AB \cong A'B'$, $AC \cong A'C'$, and $BC \cong B'C'$. We have only to prove that the angles are congruent. There is a ray AD, as in Fig. 6–69(c), from A on the opposite side of AC from B, such that $\angle DAC \cong \angle A'$. On AD, there is a point E, such that $AE \cong A'B'$.

STATEMENTS	REASONS
1. $\angle EAC \cong \angle A'$.	The ray AE was chosen this way.
2. $AE \cong A'B'$.	E was so chosen.
3. $AC \cong A'C'$.	Given.
4. $\triangle AEC \cong \triangle A'B'C'$.	S.A.S. theorem.

Now consider the segment BE and the triangles ABE and BEC, as in Fig. 6–70.

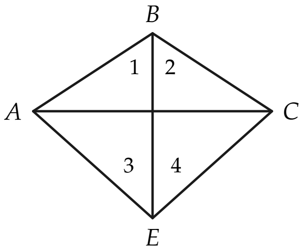


FIGURE 6–70

STATEMENTS	**REASONS**
5. $\triangle ABE$ is isosceles.	$AB \cong A'B' \cong AE$.
6. $\angle 1 \cong \angle 3$.	Statement 5 and Theorem 6–3.
7. $\angle 2 \cong \angle 4$.	Same argument as used for $\angle 1$ and $\angle 3$.
8. $\angle B \cong \angle E$.	Addition of angles postulate.
9. $\triangle ABC \cong \triangle AEC$.	S.A.S. theorem.
10. $\triangle ABC \cong \triangle A'B'C'$.	Statements 4 and 9.

Remark. In our proof, we have tacitly assumed that BE intersects the segment AC at a point between A and C. This need not be the case. We could have either Fig. 6–71(a) or 6–71(b).

Problem. Prove both cases.

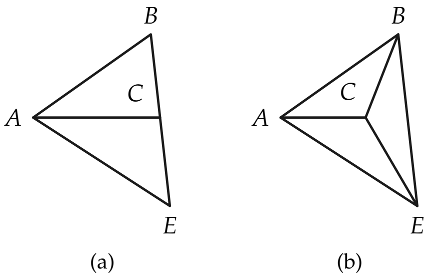


FIGURE 6–71

Exercise Group 6–11

1. You can conclude from Theorem 6–7 that a triangle is *rigid*. Why? Why is a brace put on a gate? (Fig. 6–72)

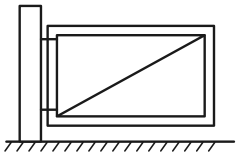


FIGURE 6–72

2. If, in the figure, $\overline{AB} = \overline{BC}$, and $\overline{AD} = \overline{DC}$, prove that $\angle ABD \cong \angle DBC$. (Fig. 6–73)

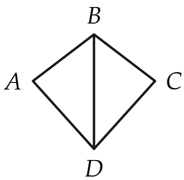


FIGURE 6–73

3. If, in the figure, $\overline{AB} = \overline{DC}$, and $AD \cong BC$, prove that $\angle CAB \cong \angle ACD$ and $\angle D = \angle B$. (Fig. 6–74)

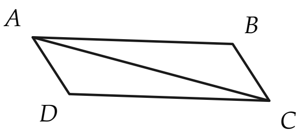


FIGURE 6–74

4. If, in the figure, $AB \cong BC$, and $AD \cong DC$, prove that BD bisects $\angle ABC$. (Fig. 6–75)

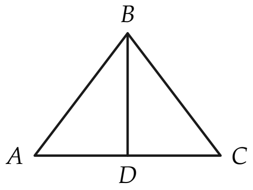


FIGURE 6–75

5. If, in the figure, $AB \cong AD$, and $BC \cong CD$, prove that $\angle 1 \cong \angle 2$, $\triangle ABE \cong \triangle ADE$, and $\angle AEB$ is a right angle. (Fig. 6–76)

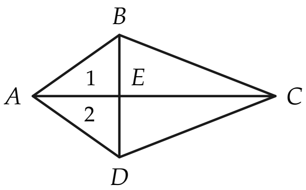


FIGURE 6–76

6. If, in the figure, $\overline{AB} = \overline{CD}$, and $\overline{BC} = \overline{AD}$, and $\overline{AF} = \overline{EC}$, prove that $\overline{BF} = \overline{ED}$. (Fig. 6–77)

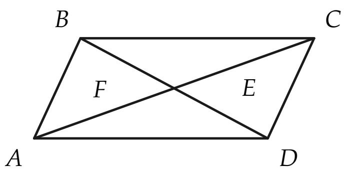


FIGURE 6–77

7. If, in the figure, $\triangle ABC$ is isosceles with vertex B, and $AD \cong EC$, prove that $BE \cong BD$. (Fig. 6–78)

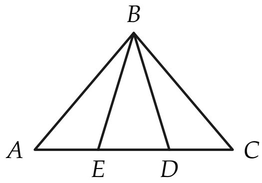


FIGURE 6–78

8. If, in the figure, $AB \cong AD$, and $BC \cong CD$, prove that $\angle B \cong \angle D$. (Fig. 6–79)

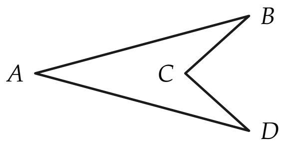


FIGURE 6–79

9. Given: $CE \cong CF$,
$AE \cong BF$,
$AD \cong BD$.
Prove: $\angle 1 \cong \angle 2$.
(Fig. 6–80)

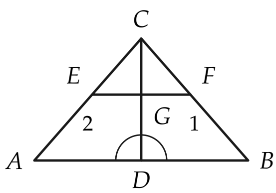


FIGURE 6–80

10. Given: $\angle 1 \cong \angle 2$,
$\angle DCE \cong \angle DCF$,
$AC \cong BC$.
Prove: $AD \cong BD$.
(Fig. 6–81)

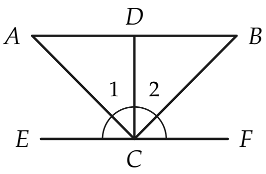


FIGURE 6–81

11. If $AC \cong BD$, and $AD \cong BC$, show that $\angle C \cong \angle D$. (Fig. 6–82)

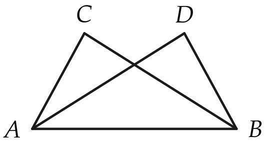


FIGURE 6–82

12. If $AB \cong BC$, and $BD \cong BE$, show that $\angle A \cong \angle C$. (Fig. 6–83)

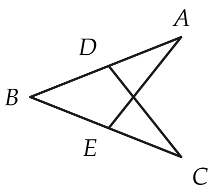


FIGURE 6–83

13. Given: $AC \cong BC$,
$AD \cong DB$.
Prove: $AO \cong BO$.
(Fig. 6–84)

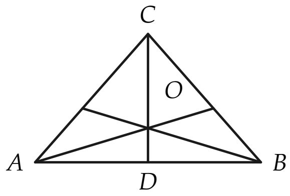


FIGURE 6–84

14. Prove that $AD \cong BE$ if $AC \cong BC$ and $\angle 1 \cong \angle 2$. (Fig. 6–85)

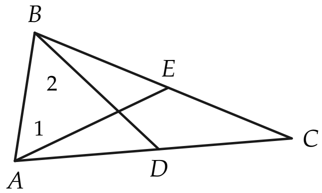


FIGURE 6–85

15. Given: $AD \cong BC$,
$AB \cong DC$,
$AM \cong MC$.
Prove: $EM \cong MF$.
(Fig. 6–86)

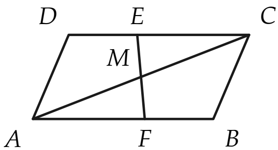


FIGURE 6–86

16. Given: $\angle 1 \cong \angle 2$,
$AE \cong DB$.
Prove: $\angle 3 \cong \angle 4$.
(Fig. 6–87)

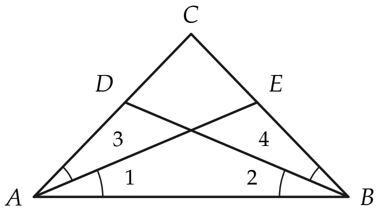


FIGURE 6–87

17. Given: $BE \cong CE$,
$\angle 1 \cong \angle 2$.
Prove: $\angle 3 \cong \angle 4$.
(Fig. 6–88)

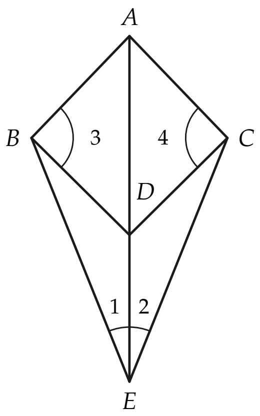


FIGURE 6–88

18. Prove that the medians from the base angles of an isosceles triangle are congruent.

19. Given: $AD \cong BC$,
$AB \cong CD$,
$\angle 1 \cong \angle 2$.
Prove: $DE \cong BF$.
(Fig. 6–89)

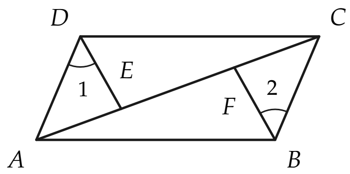


FIGURE 6–89

20. Given: $AD \cong CB$,
$BD \cong AC$.
Prove: $\triangle DEC$ is isosceles.
(Fig. 6–90)

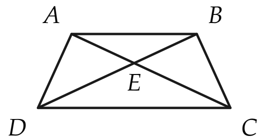


FIGURE 6–90

21. Given: $AB \cong CD$,
$AE \cong CF$,
$\angle 3 \cong \angle 4$.
Prove: $\angle 1 \cong \angle 2$.
(Fig. 6–91)

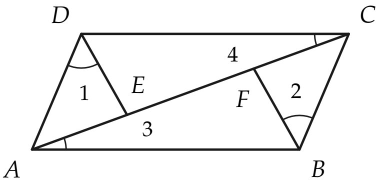


FIGURE 6–91

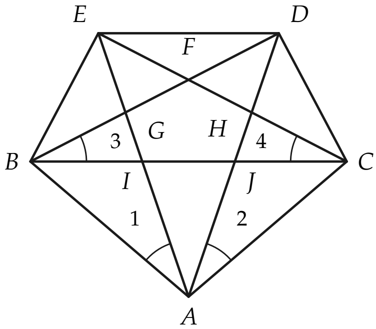


FIGURE 6–92

22. If triangles ABC and ADE are isosceles with vertex A, and $\angle 1 \cong \angle 2$, prove that $\angle 3 \cong \angle 4$, $FG \cong FH$, and $GI \cong HJ$. (Fig. 6–92)

23. Prove that the line joining the vertex of an isosceles triangle to the midpoint of the base bisects the vertex angle, and conversely.

6–10 RIGHT ANGLES

In Definition 6–1 we defined right angles, but so far *we do not know that there are such angles*. We now show how to use our congruence theorem (S.A.S.) to show that there is a right angle in our geometry.

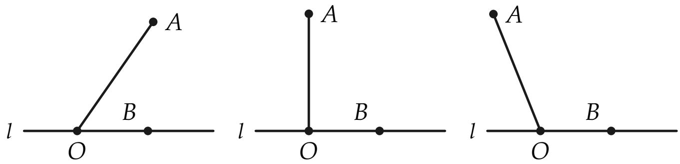


FIGURE 6–93

Proof. On a line l, choose a point O and let $\angle AOB$ be an angle (not straight) with one side on l (Fig. 6–93). Now consider the angle congruent to $\angle AOB$, with vertex O, one side OB on the line l and the other on the opposite side of l from A. Choose the point A' on this last side, such that $OA \cong OA'$. Because A and A' are on opposite sides of l, the segment AA' intersects the line in some point C. There are three cases to consider, as shown in Fig. 6–94.

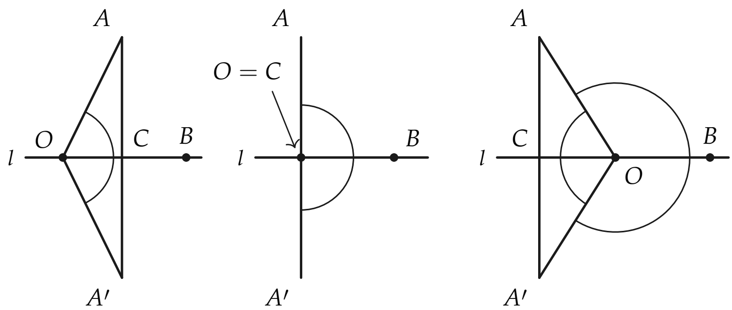


FIGURE 6–94

If C is on the same side of O as B, then by the S.A.S. theorem, $\triangle AOC \cong \triangle A'OC$. Hence $\angle OCA \cong \angle OCA'$. Since these angles are congruent and supplementary to each other, they are right angles. In case $C = O$, we have $\angle AOB \cong \angle A'OB$, and hence they are right angles. If C is on the opposite side of O from B, then the angles AOC and $A'OC$ are congruent. (Why?) We may apply the S.A.S. congruence theorem. $\angle ACO \cong \angle A'CO$, and these are right angles.

Exercise Group 6–12

1. Perform a construction to draw a right angle.
2. If, in the figure, $PQ \cong PS$, and $\angle QPR \cong \angle SPR$, prove that $\angle PRQ$ is a right angle. (Fig. 6–95)

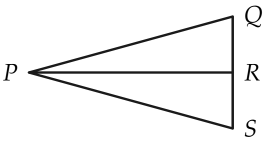


FIGURE 6–95

3. If, in the figure, $AB \cong AD$, and $BC \cong CD$, prove that $\angle AEB$ is a right angle. (Fig. 6–96)

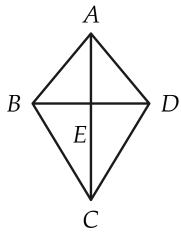


FIGURE 6–96

6–11 INEQUALITIES FOR ANGLES

Even before studying geometry you used angles of the same size and compared angles of different sizes (by saying, for example, that one angle is larger than another). In this section, we see how to compare the sizes of angles. To do this, we have to define $>$ (is greater than) and $<$ (is less than) for angles. The idea we use in forming this definition is illustrated in Fig. 6–97. If $\angle B'$ is congruent to $\angle B$, and $\angle B'$ and $\angle A$ are as drawn (that is, the other side of $\angle B'$ is *in the interior* of $\angle A$), then we would say that $\angle B < \angle A$. In Fig. 6–97, we have, so to speak, "moved" $\angle B$ to the position of $\angle B'$ for comparison. Notice how the definition below parallels the definition of "greater than" and "less than" for segments.

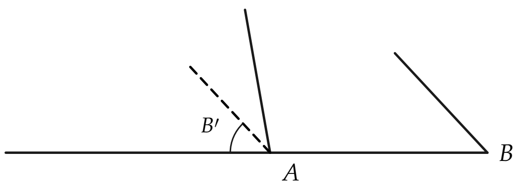


FIGURE 6–97

Definition 6–5. Let A and B be any angles not straight angles. Let $\angle B'$ be an angle congruent to $\angle B$, such that one side of $\angle B'$ is the same as the first side of $\angle A$, and the other side of $\angle B'$ and the second side of $\angle A$ both lie on the same side of the first side of $\angle A$.

(a) If the second side of $\angle B'$ is interior to $\angle A$, we say that $\angle B < \angle A$. (See Fig. 6–98.)

(b) If the second side of $\angle B'$ is exterior to $\angle A$, we say that $\angle B > \angle A$. (See Fig. 6–99.)

(c) If $\angle A$ is a straight angle, and $\angle B$ is not, we say that $\angle A > \angle B$ and that $\angle B < \angle A$.

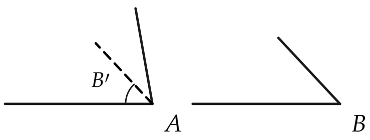


FIGURE 6–98

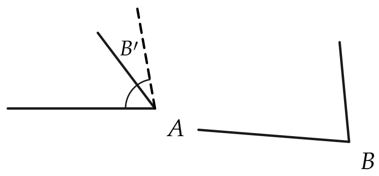


FIGURE 6–99

Our intuition tells us that if $\angle B < \angle A$, then $\angle A > \angle B$, and if $\angle B > \angle A$, then $\angle A < \angle B$. These statements require proof. Since the proofs of these statements are essentially the same, we prove only the first.

Theorem 6–8. *If $\angle B$ and $\angle A$ are two angles, such that $\angle B < \angle A$, then $\angle A > \angle B$.*

**Proof.* Figure 6–100 shows that $\angle B < \angle A$. To show that $\angle A > \angle B$, we must show that the second side of $\angle A'$ is in the exterior of $\angle B$, as in Fig. 6–101.

*This proof is difficult. If the teacher wishes, Theorem 6–8 may be treated as a postulate.

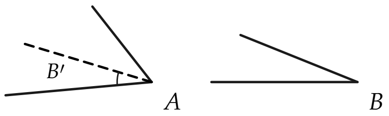


FIGURE 6–100

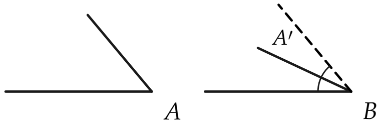


FIGURE 6–101

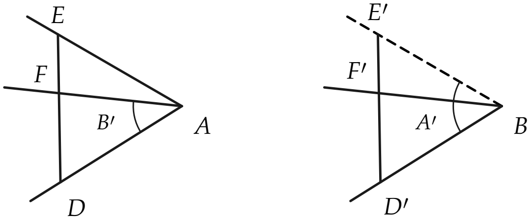


FIGURE 6–102

Choose points D and E on the sides of $\angle A$ and points D' and E' on the sides of $\angle A'$ such that $AD \cong BD'$ and $AE \cong BE'$. (See Fig. 6–102.) DE intersects the second side of $\angle B'$ in some point F between D and E because the second side of $\angle B'$ is in the interior of $\angle A$. $\triangle ADE \cong \triangle BD'E'$ by the S.A.S. theorem, and hence $DE \cong D'E'$. Choose F' between D' and E', such that $DF \cong D'F'$. This is possible because $D'E' \cong DE > DF$. The ray BF' is in the interior of $\angle A'$. (Why?) $\angle D \cong \angle D'$, and hence $\triangle FDA \cong \triangle F'D'B$. $\angle DAF \cong \angle D'BF'$, and $\angle DAF$ was given congruent to $\angle B$. By the three-part angle postulate, $\angle B \cong \angle D'BF'$; it follows that the ray BF' is the same as the second side of $\angle B$. Therefore, the second side of $\angle B$ is interior to $\angle A'$ or, what is the same, the second side of $\angle A'$ is in the exterior of $\angle B$. This completes the proof that $\angle A > \angle B$.

Remark. Theorem 6–8 tells us that we get the same result regardless of whether we "move" B to compare with A, or "move" A to compare with B.

Theorem 6–9. *For any two angles $\angle A$ and $\angle B$, either $\angle A < \angle B$, or $\angle A \cong \angle B$, or $\angle A > \angle B$; and exactly one of these is true.*

Proof. If $\angle A$ is a straight angle, either $\angle A \cong \angle B$ and $\angle B$ is also a straight angle or, by Definition 6–5, $\angle A > \angle B$. Similarly, if $\angle B$ is a straight angle, either $\angle B \cong \angle A$ or $\angle B > \angle A$. If neither $\angle A$ nor $\angle B$ is straight, then the second side of $\angle B'$ coincides with the second side of $\angle A$, and $\angle B \cong \angle A$; or the second side of $\angle B'$ is interior to $\angle A$, and $\angle A > \angle B$; or the second side of $\angle B'$ is exterior to $\angle A$, and $\angle A < \angle B$. Since only one of these possibilities can hold, the proof is complete.

Some simple properties of inequalities are collected in the following theorem.

Theorem 6–10. *(a) If $\angle A \cong \angle B$, and $\angle B < \angle C$, then $\angle A < \angle C$.*

(b) If $\angle A < \angle B$, and $\angle B \cong \angle C$, then $\angle A < \angle C$.

(c) If $\angle A \cong \angle B$, and $\angle B > \angle C$, then $\angle A > \angle C$.

(d) If $\angle A < \angle B$, and $\angle B < \angle C$, then $\angle A < \angle C$.

*(e) If $\angle A \geqq \angle B$, and $\angle B \geqq \angle C$, then $\angle A \geqq \angle C$.**

Proof of (a). Since $\angle B < \angle C$, on "moving" $\angle B$ to compare with $\angle C$, the second side of $\angle B'$ is interior to $\angle C$. Since $\angle A \cong \angle B$, the second sides of $\angle A'$ and $\angle B'$ would be the same. Hence $\angle A < \angle C$. The proofs in all the other cases are essentially the same.

Exercise Group 6–13

*The symbol $\geqq$ means "greater than or congruent to," as in Chapter 4.

1. Can you think of any other properties not included in Theorem 6–10? There are several. Write them down. Draw figures to illustrate them.
2. Draw figures to indicate that parts (b) through (e) of Theorem 6–10 are true. Be sure you understand how to use Definition 6–5.

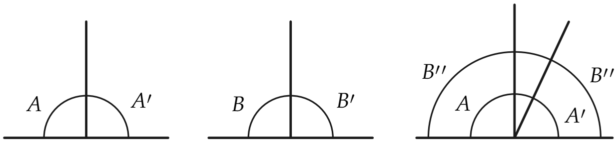
Definition 6–6. An angle less than a right angle is called an *acute* angle. An angle greater than a right angle is called an *obtuse* angle.

6–12 RIGHT ANGLES AGAIN

We can now use Theorem 6–9 to prove that all right angles are congruent to each other. It is interesting, from a historical point of view, to know that Euclid *postulated* this fact. The proof we give below could have been formulated by Euclid just as well as he formulated other proofs. Not only did Euclid ignore postulates that were needed (like the betweenness postulates), but he also introduced postulates that were not needed.

Theorem 6–11. *All right angles are congruent to each other.*

Proof. Suppose that two right angles are $\angle A$ and $\angle B$, and that their supplements are $\angle A'$ and $\angle B'$, as in Fig. 6–103(a) and 6–103(b). There are only three possibilities, according to Theorem 6–9. Either $\angle A < \angle B$, or $\angle A \cong \angle B$, or $\angle A > \angle B$, and one of these must be true. We will show that $\angle A < \angle B$ and $\angle A > \angle B$ are false. Suppose that $\angle A < \angle B$, and that $\angle B''$ is congruent to $\angle B$ and located as in Fig. 6–103(c). Let B''' be the supplement of $\angle B''$.



(a) (b) (c)

FIGURE 6–103

STATEMENTS	**REASONS**
1. $\angle B'' \cong \angle B$.	By definition of $\angle B''$.
2. $\angle B'''$ is the supplement of $\angle B''$, and $\angle B'$ is the supplement of $\angle B$.	By definition of supplement.
3. $\angle B' \cong \angle B'''$.	If angles are congruent, their supplements are congruent.
4. $\angle B \cong \angle B'$.	$\angle B$ is a right angle.
5. $\angle B \cong \angle B'''$.	Angle congruence is transitive.
6. $\angle A < \angle B$.	Hypothesis.
7. $\angle A < \angle B'''$.	$\angle A < \angle B \cong \angle B'''$.
8. $\angle B''' < \angle A'$.	The "second" side of $\angle B'''$ is in the interior of $\angle A'$.
9. $\angle B''' < \angle A$.	$\angle B''' < \angle A' \cong \angle A$.
10. $\angle A < \angle A$.	Statements 7 and 9.

Since $\angle A \cong \angle A$, we know by Theorem 6–9 that it is impossible for $\angle A$ to be less than itself. Therefore, it is impossible that $\angle A < \angle B$. In the same way, it can be shown that it is impossible that $\angle B < \angle A$. Therefore, we must have $\angle A \cong \angle B$. This completes the proof.

Definition 6–7. Two lines, l and m, are said to be *perpendicular* to each other if they intersect and the angles formed are right angles, as in Fig. 6–104(a). To indicate that lines are perpendicular we write "$l \perp m$."

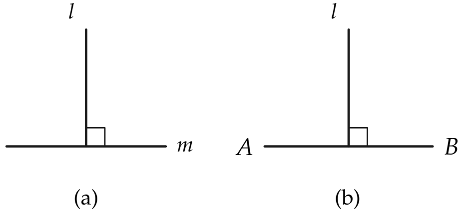


FIGURE 6–104

The line l is called the *perpendicular bisector* of the segment AB if $l \perp AB$ and l passes through the midpoint of AB, as in Fig. 6–104(b). This is written "$l \perp$ bis AB," and read "l is the perpendicular bisector of AB."

Exercise Group 6–14

1. Prove the theorem that if the two legs of one right triangle are congruent, respectively, to the two legs of another right triangle, then the triangles are congruent.
2. In the figure, angles ADC and BCD are right, $AD \cong BC$, and $AB \cong CD$. Prove that $\triangle ADC \cong \triangle BCD$ and $\triangle ABD \cong \triangle BAC$. (Fig. 6–105)

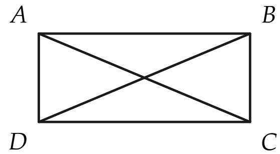


FIGURE 6–105

3. If, in the figure, $AD \cong DC$, and $BD \perp AC$, prove that $\angle BAD \cong \angle BCD$. (Fig. 6–106)

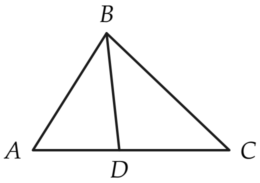


FIGURE 6–106

4. If, in the figure, $AD \cong BC$, angles ADC and BCD are right, and $DE \cong EC$, prove that $AE \cong BE$. (Fig. 6–107)

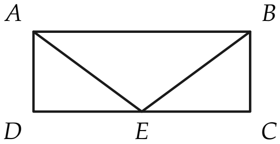


FIGURE 6–107

5. Given: $BD \perp AC$,
   $\angle ABD \cong \angle DBC$.
   Prove: $\angle A \cong \angle C$.
   (Fig. 6–108)

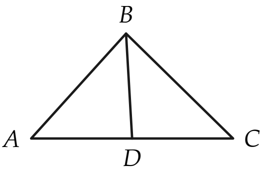


FIGURE 6–108

6. Using Fig. 6–108, prove that $\angle A \cong \angle C$ if $BD \perp$ bis AC.
7. In the figure, $AB \perp BD$, $DE \perp BD$, and $BC \cong CD$. Prove that $AB \cong DE$. (Fig. 6–109)

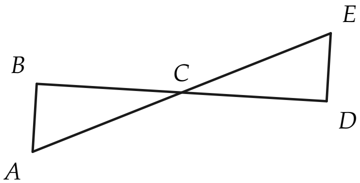


FIGURE 6–109

8. Given: $AB \perp DC$,
   $BE \cong BC$,
   $\angle DEB \cong \angle C$.
   Prove: $DE \cong EC$.
   (Fig. 6–110)

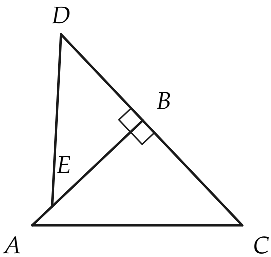


FIGURE 6–110

9. Prove that the bisector of the vertex angle of an isosceles triangle is the perpendicular bisector of the base.
10. Prove that the median from the vertex of an isosceles triangle is perpendicular to the base.
11. Given: $AC \cong AD$,
    $AC \perp CB$,
    $AD \perp DB$.
    Prove: $\triangle ACB \cong \triangle ADB$.
    (Fig. 6–111)

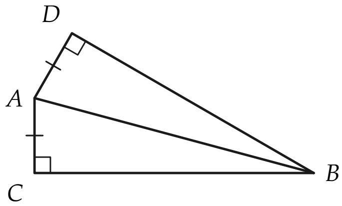


FIGURE 6–111

REVIEW OF CHAPTER 6

In this chapter there are many important ideas and geometric facts. It is most important that you review the topics listed below. Each topic contains many ideas, so study each one separately. Then study the whole list to be sure that you understand how the topics are related.

1. First in importance are the four postulates of Group V. It is not necessary to memorize the postulates word for word, but you must understand the meaning of each one so that you will know when you can use it.

2. We also gave several important definitions. Be sure that you know definitions for the following: right angle, congruent triangles, $>$ and $<$ for angles, acute angle, obtuse angle.

3. Below we list the most important theorems proved in this chapter. You should be able to prove many of them. In reviewing the proofs of these theorems, first see whether you can formulate a proof without looking at the text.

> If two angles are congruent, their supplements are congruent.
> Vertical angles are congruent.
> In an isosceles triangle, the base angles are congruent.
> The first congruence theorem (S.A.S.).

For any two angles $\angle A$ and $\angle B$, exactly one of the following is true: $\angle A > \angle B$, $\angle A \cong \angle B$, $\angle A < \angle B$.
If $\angle A \cong \angle B$, and $\angle B < \angle C$, then $\angle A < \angle C$, etc.
All right angles are congruent.
The second congruence theorem (A.S.A.).
The third congruence theorem (S.S.S.).

Review Exercises

1. Draw an $\angle A$. Draw a ray r from a point O. Draw the two angles congruent to A that have the ray r as one side. See Postulate V–1.

2. If $\angle A \cong \angle D$, and $\angle D \cong \angle E$, and $\angle E \cong \angle F$, prove that $\angle A \cong \angle F$. See Postulate V–2.

3. If in Fig. 6–112, $\angle A_1 \cong \angle R$, and $\angle A_2 \cong \angle S$, prove that $\angle B \cong \angle T$. See Postulate V–3.

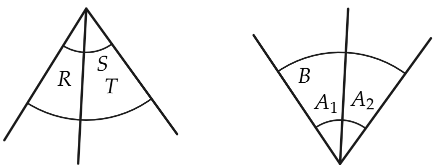


FIGURE 6–112

4. If in Fig. 6–113, $\angle A \cong \angle B$, prove that $\angle C \cong \angle D$. What is the converse of this? Is the converse true? What is the contrapos-

itive of this? Is it true?

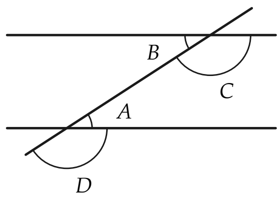


FIGURE 6–113

5. If in Fig. 6–114, $\angle AOB \cong \angle BOC$, what is $\angle AOB$ called?

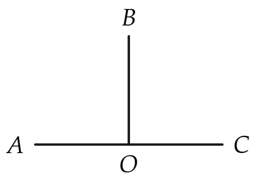


FIGURE 6–114

6. If (Fig. 6–115) $FE \cong ST$, $EG \cong RT$, and $\angle E \cong \angle T$, to what angle is $\angle F$ congruent?

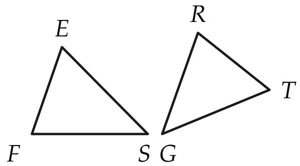


FIGURE 6–115

7. If (Fig. 6–116) $\overline{RS} = \overline{TS}$, prove that $\angle R \cong \angle T$.

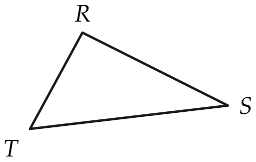


FIGURE 6–116

8. Using Fig. 6–115, prove that the triangles are congruent.

9. Figure 6–117 shows a way to measure the width $\overline{PQ}$ of a river. From any point R not on the line PQ, RU is sighted so that $\angle URP \cong \angle PRQ$. From P a line PT is sighted so that $\angle TPR \cong \angle QPR$. The point S, where PT and RU intersect, is determined, and PS is measured. Why is this the distance from P to Q?

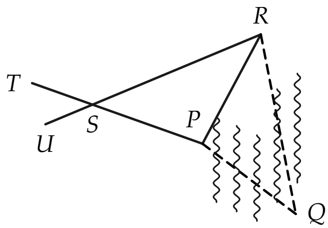


FIGURE 6–117

10. The first congruence theorem says that if two sides and the included angle of one triangle are congruent to two sides and the included angle of another triangle, then the triangles are congruent. What is the converse of this theorem? Is it true?

11. If a triangle is equilateral, then it is isosceles. What is the converse of this statement? Is it true? What is the contrapositive? Is it true?

12. If in Fig. 6–118, $\triangle ABC$ and $\triangle ADC$ are isosceles, with vertices B and D, prove that BD bisects $\angle ABC$.

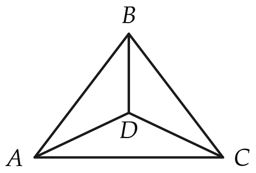


FIGURE 6–118

13. If (Fig. 6–119) $\triangle ABC$ is isosceles, and $\angle ABD \cong \angle EBC$, prove that $\overline{AD} = \overline{EC}$.

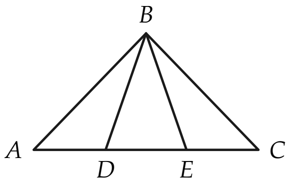


FIGURE 6–119

14. If (Fig. 6–120) $\overline{AB} = \overline{BC}$, $\overline{AD} = \overline{DC}$, and $\angle ADE \cong \angle CDF$, prove that $BE \cong BF$.

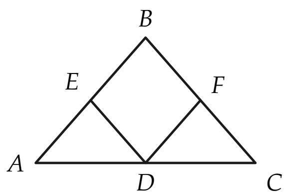


FIGURE 6–120

15. When we build a house, the sub-flooring and the sheathing boards are usually laid diagonally. Why?

16. If (Fig. 6–121) $QS \cong SR$, and $PS \perp QR$, prove that $PQ \cong PR$.

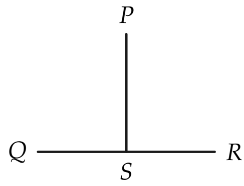


FIGURE 6–121

17. If the point O bisects the segments AB and CD, prove that $\angle A \cong \angle B$. See Fig. 6–122.

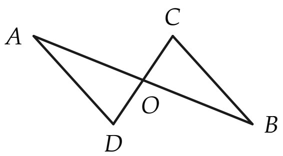


FIGURE 6–122

18. Show that if two isosceles triangles have congruent vertex angles, and their congruent sides have the same length, then the triangles are congruent.

19. If in Fig. 6–123, $AB \cong AD$, $BC \cong DE$, and $AC \cong AE$, to what angle is $\angle B$ congruent? Why?

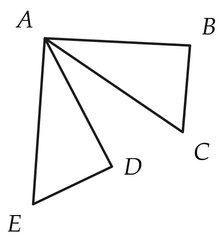


FIGURE 6–123

20. If in Fig. 6–124 $\triangle ABC$ is equilateral, and D, E, F bisect the sides, prove that $\triangle EFD$ is equilateral.

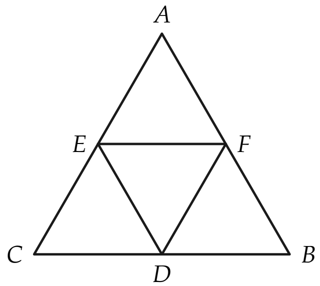


FIGURE 6–124

21. If (Fig. 6–125) $\angle CAD \cong \angle BDA$, and $\angle CDA \cong \angle BAD$, prove that $\overline{AB} = \overline{CD}$.

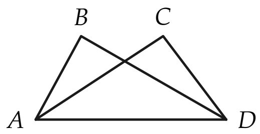


FIGURE 6–125

22. Prove that if two points are equidistant from the ends of a segment, they determine the perpendicular bisector of the segment.

23. Prove that if the bisector of an angle of a triangle is perpendicular to the opposite side, then it bisects that side, and the triangle is isosceles.

24. Prove that any point on the perpendicular bisector of a segment is equidistant from the ends of the segment.

25. Given: $\angle 1 \cong \angle 2$,
$AC \cong BC$,
$IJ \cong IK$.
Prove: $GJ \cong HK$,
$GF \cong FH$.
(Fig. 6–126)

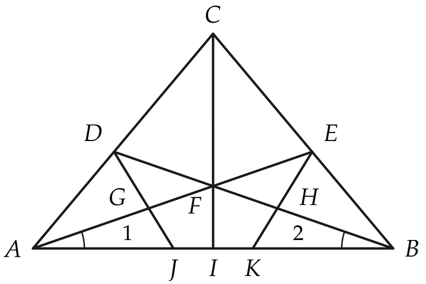


FIGURE 6–126

Test 1

True-False

1. If a theorem is true, its contrapositive is true.
2. The supplement of a 40° angle is a 50° angle.
3. If $\angle A < \angle B$, and $\angle B < \angle C$, and C is a 50° angle, then $\angle A$ is less than a 49° angle.
4. If ABC (B is between A and C), DEF, $AC \cong DF$, and $AB > DE$, then $BC > EF$.
5. If $\triangle ABC$ is isosceles, with vertex A, and D, E are the mid-points of AB, AC respectively, then $BE \cong CD$.
6. If in triangles ABC and $A'B'C'$, $\angle A \cong \angle A'$, $\angle B \cong \angle B'$, $\angle C \cong \angle C'$, and $AB \cong A'C'$, then $\triangle ABC \cong \triangle A'B'C'$.

7. If $\angle ABC$ is 40°, and $\angle DBC$ is 50°, then $\angle DBA$ is a right angle.
8. If $\angle ABC$ is 60°, and $\angle DBC$ is 120°, then either $\angle DBA \cong \angle ABC$, or $\angle DBA$ is a straight angle.
9. If segments AB and CD intersect at O, then $\angle AOB \cong \angle BOD$.
10. For all statements α, β, γ, such that $\alpha \to \beta$ is true and $\beta \to \gamma$ is false, the implication $\alpha \to \gamma$ is false.

Test 2

1. Lines l and m intersect so that a pair of adjacent angles differ by 18°. Determine the number of degrees in each angle.
2. If $\overline{AB} = 24$, and ACB, and $\overline{AC} = 3 \cdot \overline{CB}$, compute $\overline{CB}$.
3. If $\overline{AB} = 24$, ABC, and $\overline{AC} = 3 \cdot \overline{CB}$, compute $\overline{CB}$.
4. Line segments AB and CD are congruent. The length of AB as measured by EF is 12. The length of CD as measured by GH is 36. What is the length of GH as measured by EF?
5. The length of AB as measured by CD is $\frac{3}{4}$. What is the length of CD as measured by AB?
6. Let AB be a unit segment. Designate the midpoint of AB by A_1, the midpoint of AA_1 by A_2, the midpoint of AA_2 by A_3, etc. What is the length of A_5A_3?

Algebra Review

1. The perimeter of an isosceles triangle is 75 ft. One side is 15 ft longer than the second side. Prove that the shortest side of the triangle is either 15 ft long or 20 ft long.
2. A triangle has two equal angles. The length of the included side is 7 in. The perimeter of the triangle is 43 in. Determine the lengths of the other two sides of the triangle.

3. $\triangle ABC$ is equilateral. Points X, Y, and Z are the midpoints of sides AB, BC, and CA, respectively. The perimeter of $\triangle XYZ$ is 30 in. Determine the length of each side of $\triangle ABC$.

4. In Fig. 6–127, $\overline{AE} = \overline{AD}$, and $\overline{AB} = \overline{AC}$. $\overline{BE} = 18$ in., and $\overline{DO} = \frac{1}{2}\overline{OC}$. Determine the length of segment OC.

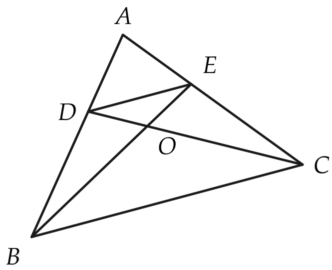


FIGURE 6–127

5. In $\triangle ABC$, the bisector of vertex angle A is perpendicular to the base BC at point D. $\overline{BD} = \overline{CD}$. $\overline{AB} - \overline{BD} = 8$ in., and the perimeter of $\triangle ABC$ is 40 in. Determine the length of each side of $\triangle ABC$.

6. Prove that if $(x+2)(x-3) = 0$, and x is positive, then $x = 3$.

7. Prove that if $(x+2)(x-3) = 0$, and x is negative, then $x = -2$.

8. A businessman bought several suits of clothes. He paid \$40 each for some of the suits and \$50 each for the rest. If he paid a total of \$1500 for these suits, prove that he could not possibly have bought 22 of the \$40 suits.

*9. Prove that the following problem has no solution: Find two numbers whose sum is 32, such that if the larger of the numbers is divided by 6, and the smaller is divided by 5, then the sum of the quotients is 6.

*10. Describe all possible pairs of numbers a and b such that $(a-2)/(2b-9) = a-2$.